Type systems: scales, line-height, measure

Week 3, Lecture 2, Design for Builders: Ship Beautiful Products as a Founder

Autumn 2026

What we will cover

- Why a modular type scale produces consistent size relationships without guessing
- How to set line-height as a ratio, not a fixed pixel value
- The 45-75ch measure rule and why column width controls reading comfort
- A repeatable strategy for pairing two typefaces without visual chaos

Part 1: the type scale

Why arbitrary sizes look amateur

Most builder products have font sizes like 13, 16, 20, 24, 32, 48. Those sizes are not bad. But they were probably chosen one at a time, each in isolation.

The result: no consistent visual relationship between sizes. Readers cannot tell what level of hierarchy they are at, because the steps between sizes are uneven (Figma 2023).

A **modular type scale** solves this by deriving every size from a single base and ratio.

How a modular scale works

Choose a base size and a ratio. Multiply up and divide down:

```
Base: 16px
Ratio: 1.25 (Major Third)

xs:   16 ÷ 1.25 ÷ 1.25 = 10.24px  (~10px)
sm:   16 ÷ 1.25        = 12.8px   (~13px)
base: 16px
lg:   16 × 1.25        = 20px
xl:   16 × 1.25 × 1.25 = 25px
2xl:  16 × 1.25^3      = 31.25px  (~31px)
```

Every size relates to every other by the same ratio. That relationship is what makes the scale look intentional (Figma 2023).

Common scale ratios and when to use them

Ratio	Name	Typical use
1.125	Major Second	Dense UIs, data tables
1.25	Major Third	Product dashboards, apps
1.333	Perfect Fourth	Marketing pages, blogs
1.5	Perfect Fifth	Editorial, large type, posters
2.0	Octave	Headlines only, no body

Most product UIs use 1.25. Marketing and landing pages often use 1.333. Using a ratio above 1.5 for a full scale produces sizes that are too spread apart for a five-step system.

Five steps is usually enough

A five-step scale covers:

- Label / caption (xs or sm)
- Body text (base)
- Card title or subhead (lg)
- Section heading (xl)
- Page headline (2xl)

If you have more than five distinct size levels in a product, you probably have a hierarchy problem, not a scale problem. Adding more sizes does not fix unclear hierarchy.

Registering the scale as Figma text styles

Once you have the five sizes, create a Figma text style for each (Figma 2023):

- Name them by role, not pixel value: `text/body`, `text/heading-sm`, `text/heading-lg`
- Not: `text/16px`, `text/24px`

Role-based names survive a scale change. If you decide the base should be 15px, you change one value and re-derive. Pixel-named styles break on every change.

Part 2: line-height and measure

The problem with “always use 1.5”

Line-height 1.5 is the browser default recommendation. It is everywhere. It is also wrong for most situations (Butterick 2019).

The correct range for body text is **120-145% of point size** (line-height 1.2 to 1.45 in unitless CSS notation). 1.5 is above that range for most body sizes. It wastes vertical space and makes the column feel airy in a way that reads as unfinished.

For headlines at 36px or above, 1.5 is not just wrong: it looks like a broken layout.

Line-height is proportional to size

Wathan and Schoger (2019) add an important nuance: line-height scales with font size in the opposite direction.

```
Small text (12–14px):  needs more leading. Use 1.5–1.7
Body text  (15–18px):  1.35–1.5
Large text (24–32px):  1.15–1.3
Headlines  (36px+):    1.0–1.15
```

A 48px headline with line-height 1.5 looks like a double-spaced essay. A 13px caption with line-height 1.1 looks crushed. Match the ratio to the size.

Measure: the 45-75ch rule

Measure is the width of a line of text, measured in characters.

- Below 45 characters per line: too narrow. Eyes make too many jumps. Reading pace drops.
- 45-75 characters: comfortable reading range. Most prose reads well here.
- Above 75 characters: the eye struggles to track from the end of one line back to the start of the next. Reading fatigue increases.

In CSS, `max-width: 65ch` on a `p` element constrains body text to approximately 65 characters per line, regardless of viewport width (Butterick 2019).

Line-height and measure interact

Wathan and Schoger (2019) show that a narrow column (35ch) can tolerate a tighter line-height (1.3) because the short line length gives the eye a natural rhythm. A wide column (70ch) needs more leading (1.5-1.6) to help the eye track from line end back to line start.

The practical rule: set measure first, then tune line-height to match.

```
Narrow column (~40ch):   line-height 1.3–1.4
Standard column (~60ch): line-height 1.4–1.5
Wide column (~70ch):     line-height 1.5–1.6
```

Part 3: pairing two typefaces

Why pair at all?

One well-chosen typeface used consistently is better than two poorly chosen typefaces used everywhere.

But a two-typeface system is useful when you need to distinguish between different types of content: long-form body text and short display headlines often work better with different faces. The body face optimizes for readability at 16px. The display face optimizes for impact at 36px.

The risk: clash. Two typefaces that compete instead of cooperate produce visual noise (Segall 2024).

The contrast principle for pairing

Pairs that work share one characteristic but contrast on another.

What to contrast:

- Classification: serif headline with sans-serif body (or vice versa)
- Personality: a formal serif with a neutral neo-grotesque
- Weight range: a typeface with a wide weight range paired with one that has a narrow range

What to share:

- X-height: two typefaces with similar x-heights look like they belong together at the same size
- Era or origin: both humanist, or both geometric, even if one is serif and one is sans

The one rule that prevents most pairing mistakes

Do not pair two typefaces from the same classification and the same personality.

- Two geometric sans-serifs: they look like near-duplicates, not a system
- Two high-contrast serifs: the page looks indecisive, not designed
- A grotesque sans body with a slightly different grotesque sans headline: readers cannot tell why there are two typefaces

When you cannot see a clear reason for the pairing from 50 centimeters away, it is the wrong pair (Segall 2024).

A worked example: pairing for a SaaS product

Product: a project management tool for engineering teams. Adjectives: professional, clear, modern.

Body face: Inter (neo-grotesque, neutral, excellent legibility at 15-16px, wide weight range).

Display face option A: Playfair Display (high-contrast transitional serif). Contrast: strong. Shared: both have larger x-heights than their older counterparts. Problem: Playfair's decorative letterforms feel too editorial for an engineering tool.

Display face option B: IBM Plex Sans (humanist sans, slightly more personality than Inter, designed for technical documents). The contrast is lower, but both faces come from the same technical-professional register. They look like they belong together.

Choose option B. The product is for engineers, not magazine readers.

Take-aways

1. A modular type scale derives every size from one base and one ratio. Five steps cover all the hierarchy a typical product needs.
2. Name Figma text styles by role (`text/body`) not by pixel value. Role-based names survive scale changes.
3. Line-height is not 1.5 everywhere. It scales inversely with font size: smaller text needs more leading, larger text needs less.
4. Set body text measure to 45-75ch. The CSS property `max-width: 65ch` does this in one line.
5. Typeface pairs work when they contrast on classification or personality and share x-height or era. Two faces from the same category create confusion, not hierarchy.

> “Line-height should be thought of as proportional to the line length: short lines need less leading, long lines need more.”
>
> Adam Wathan & Steve Schoger, “Line-height is proportional,” Refactoring UI 2019

What’s next

- **This week, section:** Typography audit of your own landing page or product
- **HW2 assigned this week:** Type-only redesign. Rebuild a competitor’s landing page using only Inter, color, and spacing. Due Week 5.
- **Reading:** Week 3 reading at `/c/design-for-builders-26au/readings/wk03`
- **Week 4:** Color: building a palette you can actually use